Supplemental Material for: Adaptive Phase-Switching for Communication-Efficient Federated LoRA Fine-Tuning

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***Abstract*—This supplemental material provides hyperparameter details, per-seed results, statistical tests, a reproducibility checklist, and cross-backend verification for the main paper.**

APPENDIX

All experiments use AdamW [1] with learning rate 10^{-4}, β_1=0.9, β_2=0.999. No learning rate schedule is applied. Gradients are clipped to a maximum ℓ_2 norm of 1.0 at every optimizer step. Maximum sequence length is 256 tokens. See Table **??** for the complete configuration including ReverseAdaptive defaults.

Tables in this appendix are labelled with their corpus. Per-seed results backing a method comparison use the CUDA corpus, matching the main results tables; per-seed results documenting partition sensitivity retain the larger MPS seed set, where five seeds at α=0.1 provide better evidence than three.

At α=0.5, mean final loss is statistically indistinguishable from IID for every method, with differences of 0.001 to 0.007 against per-seed standard deviations near 0.02 over three seeds. Heterogeneity manifests instead as sensitivity to the partition draw: seed-to-seed standard deviation rises by roughly an order of magnitude, from 0.5 to 1.6 $\times$ 10^{-3} under IID to 18 to 25 $\times$ 10^{-3} at α=0.5, which is also why the per-seed spread at α=0.1 in Table II is wide.

Paired t-tests over three seeds on final-round training loss and on held-out instruction-following loss. TinyLlama-1.1B results use the CUDA corpus, matching Table **??**; LLaMA-3.2-3B results use the MPS corpus.

With three seeds these tests have very low power, so effect sizes should be read alongside the p-values. No multiplicity correction is applied in the main results tables. Applying a Bonferroni correction across the five TinyLlama comparisons gives a threshold of 0.01, which the ReverseAdaptive versus Two-Phase K=8 comparison on held-out loss (p = 0.026) does not meet. That comparison is secondary under the frontier framing of Section **??**; the load-bearing comparison is ReverseAdaptive versus FFA-LoRA, which holds at $p < 10^{-3}$ on both metrics with a difference roughly twenty times the largest per-method standard deviation.

The LLaMA-3.2-3B comparison between ReverseAdaptive and Two-Phase K=8 has a 95% confidence interval of $[-0.0114, 0.0114]$, which contains the 0.0043 difference observed between the same methods at TinyLlama-1.1B. The test therefore cannot distinguish equivalence from a difference of the magnitude seen at the smaller scale.

TABLE I
PER-SEED TINYLLAMA-1.1B IID RESULTS, CUDA CORPUS, BACKING TABLE **??**. COMMUNICATION IS DETERMINISTIC ACROSS SEEDS. † FOR TWO-PHASE THE COLUMN SHOWS THE PREDETERMINED PHASE BOUNDARY K, NOT AN ADAPTIVE SWITCH EVENT; FOR FFA-LORA, A IS FROZEN FROM INITIALIZATION; FOR REVERSEADAPTIVE IT SHOWS THE ROUND AT WHICH THE PLATEAU SIGNAL TRIGGERED.

Method	Seed	Final loss	Total MB	Switch / boundary†
FLoRA	42	1.261408	2578.13	–
FLoRA	123	1.260479	2578.13	–
FLoRA	456	1.260500	2578.13	–
FedIT	42	1.260767	2578.13	–
FedIT	123	1.258420	2578.13	–
FedIT	456	1.261503	2578.13	–
Two-Phase K=8	42	1.271276	1863.13	K=8†
Two-Phase K=8	123	1.270130	1863.13	K=8†
Two-Phase K=8	456	1.270237	1863.13	K=8†
ReverseAdaptive (τ=0.01)	42	1.275514	1533.13	6
ReverseAdaptive (τ=0.01)	123	1.274610	1533.13	6
ReverseAdaptive (τ=0.01)	456	1.274481	1533.13	6
FFA-LoRA	42	1.303975	983.13	init†
FFA-LoRA	123	1.301738	983.13	init†
FFA-LoRA	456	1.303628	983.13	init†

Code: Full source code is publicly released, including all configuration YAML files, training scripts, evaluation scripts, and scripts to reproduce every figure and table from the analysis CSVs.

Seeds: All seeds listed in Table **??**. Per-seed results are in Appendix A.

Hardware: The primary corpus of 34 runs was produced on a single Apple M4 Pro workstation with 48 GB unified memory using the MPS backend, approximately 285 hours of compute, with no institutional cluster. Those runs were re-executed on commercially rented NVIDIA hardware, an RTX 4090 for TinyLlama-1.1B and an A100 40 GB for LLaMA-3.2-3B; the comparison is in Appendix A. The Dolly-15k replication and the FFA-LoRA and FedIT baselines were produced on the rented hardware only.

Data: Alpaca [2] via `tatsu-lab/alpaca` on HuggingFace, 3000-sample subset, and Dolly-15k [3] via `databricks/databricks-dolly-15k`, matched subset. Adapter checkpoints are not released; reproduction scripts are sufficient.

The primary corpus was produced on the MPS backend and

TABLE II
NON-IID RESULTS, MPS CORPUS, FIVE SEEDS AT α=0.1. THIS TABLE DOCUMENTS PARTITION SENSITIVITY RATHER THAN COMPARING METHODS, SO THE LARGER MPS SEED SET IS RETAINED; TABLE I FOLLOWS THE CUDA CORPUS BECAUSE IT BACKS A METHOD COMPARISON. SEEDS 42 AND 1000 AT α=0.1 HAD 9 AND 8 ACTIVE CLIENTS DUE TO EMPTY DIRICHLET PARTITIONS, WHICH SCALES COMMUNICATION TO 9/10 AND 8/10 OF THE FULL VALUE (1379.81 AND 1226.50 MB AGAINST 1533.13).

Setting	Seed	Final loss	Total MB	Switch round	Active clients
ReverseAdaptive, α=0.5	42	1.285933	1533.13	6	10
ReverseAdaptive, α=0.5	123	1.282625	1533.13	6	10
ReverseAdaptive, α=0.5	456	1.252133	1533.13	6	10
ReverseAdaptive, α=0.1	42	1.268353	1379.81	6	9
ReverseAdaptive, α=0.1	123	1.288985	1533.13	6	10
ReverseAdaptive, α=0.1	456	1.348083	1533.13	6	10
ReverseAdaptive, α=0.1	789	1.259200	1533.13	6	10
ReverseAdaptive, α=0.1	1000	1.273046	1226.50	6	8

TABLE III
PAIRED t-TESTS, THREE SEEDS. TINYLLAMA-1.1B ROWS USE THE CUDA CORPUS; LLAMA-3.2-3B ROWS USE MPS. POSITIVE Δ INDICATES THE FIRST METHOD HAS HIGHER LOSS.

Comparison	Δ final loss	p	Δ held-out	p
TinyLlama-1.1B, Alpaca, IID				
Two-Phase K=8 vs. FLoRA	+0.0098	4.2×10^{-5}	+0.0034	7.9×10^{-5}
ReverseAdaptive vs. FLoRA	+0.0141	1.1×10^{-5}	+0.0063	6.2×10^{-3}
ReverseAdaptive vs. Two-Phase K=8	+0.0043	3.4×10^{-4}	+0.0029	2.6×10^{-3}
FFA-LoRA vs. ReverseAdaptive	+0.0282	4.4×10^{-4}	+0.0182	3.8×10^{-4}
FFA-LoRA vs. FLoRA	+0.0423	1.7×10^{-4}	+0.0245	1.1×10^{-4}
FedIT vs. FLoRA	−0.0006	0.588	+0.0002	0.425
TinyLlama-1.1B, Dolly-15k, IID				
Two-Phase K=8 vs. FLoRA	+0.0099	7.8×10^{-5}	+0.0043	2.4×10^{-3}
ReverseAdaptive vs. FLoRA	+0.0142	2.5×10^{-5}	+0.0061	7.6×10^{-3}
LLaMA-3.2-3B, Alpaca, IID				
Two-Phase K=8 vs. FLoRA	+0.0214	3.2×10^{-4}	–	–
ReverseAdaptive vs. FLoRA	+0.0215	1.3×10^{-2}	–	–
ReverseAdaptive vs. Two-Phase K=8	+0.000012	0.997	–	–

subsequently re-executed on CUDA. Table IV summarizes the comparison; Table V details every disagreement.

Tolerances are setting-aware. IID runs require per-round training loss to agree within 0.01 absolute. Non-IID runs allow 0.025, since heterogeneous partitions amplify per-round variation, but additionally require the switch round to match exactly and permit no more than two consecutive rounds exceeding 0.01. Communication must agree within 0.01 in all settings, which is effectively exact since totals are protocol-deterministic.

Of 34 run pairs, 31 have a valid MPS reference. The three Two-Phase K=8 runs at α=0.5 do not: their MPS executions predate the bidirectional B-only implementation and never switched, recording 2578.13 MB, the full FLoRA volume. Those are absent references rather than disagreements, and CUDA is the source of truth for that configuration. Of the 31 comparable pairs, 27 agree within tolerance and 4 do not.

All four disagreements are the same phenomenon. Total communication is a step function of the discrete switch round

TABLE IV
CROSS-BACKEND VERIFICATION SUMMARY, MPS AGAINST CUDA, 34 RUN PAIRS.

Category	Runs
Agreed within setting-aware tolerance	27
Disagreed	4
Comparable pairs	31
No valid MPS reference (CUDA reported)	3
Total	34

with 110 MB granularity (Section ??), and every observed cross-backend communication difference is exactly one step: the MPS and CUDA totals in Table V are adjacent points on the same lattice. The backends do not disagree about the protocol, only about which round the plateau criterion fires, and only where that decision sits near a boundary. Two of the four are IID runs at tight thresholds (τ=0.001 and τ=0.002), where the criterion is nearly satisfied in either of two adjacent rounds; two are α=0.1 runs, where heterogeneity produces a noisier loss trajectory. This is boundary sensitivity rather than heterogeneity sensitivity.

Run 17 was investigated separately. Re-executing on CUDA with the data partition seed fixed at 456 and the training seed changed to 999 reproduced switch round 7 exactly, establishing that the one-round difference is a stable property of the backend and partition rather than run-to-run stochasticity. At α=0.1, cross-backend agreement on the discrete switch round is therefore not guaranteed: one of three partitions fired one round later on CUDA, and CUDA is reported as primary for that setting.

The switch round matched exactly for every IID run at τ=0.01 and for all three LLaMA-3.2-3B runs, which is what supports the cross-scale transfer claim in Section ??.

Byte accounting. Communication is instrumented at the transport layer, counting `numel` times `element_size` on the tensors actually transmitted, rather than derived from parameter counts. The `get_communication_cost` helper in the FFA-LoRA aggregator is not used by this accounting.

Code revisions. Runs were produced across multiple code revisions with uncommitted working-tree modifications, so no reported number is recoverable by checking out a single commit. Reproducibility rests instead on the cross-backend replication documented here. The byte accounting is revision-invariant by direct measurement: FLoRA, Two-Phase K=8 and ReverseAdaptive were each executed under two different revisions within the CUDA corpus and produced identical totals of 2578.13, 1863.13 and 1533.13 MB. Loss values carry revision uncertainty; the FLoRA and FedIT rows of Table ?? span that boundary and differ by 0.0006, against a ReverseAdaptive-to-FFA-LoRA separation of 0.0282.

REFERENCES

[1] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.

TABLE V
CROSS-BACKEND DISAGREEMENTS. EVERY COMMUNICATION DIFFERENCE IS EXACTLY ONE 110 MB STEP OF THE SWITCH-ROUND LATTICE. RUN 16 BREACHED ONLY THE CONSECUTIVE-SPIKE RULE, WITH NO COMMUNICATION OR SWITCH-ROUND DIFFERENCE.

Run	Configuration	MPS (MB)	CUDA (MB)	Failing check
11	τ=0.001, IID	2083.13	1973.13	communication, one step (s=11 vs 10)
12	τ=0.002, IID	1863.13	1753.13	communication, one step (s=9 vs 8)
16	α=0.1, seed 123	1533.13	1533.13	3 consecutive rounds $\lvert\Delta\rvert > 0.01$
17	α=0.1, seed 456	1533.13	1643.13	communication, switch round 6 vs 7, loss

[2] R. Taori, I. Gulrajani, T. Zhang, Y. Dubois, X. Li, C. Guestrin, P. Liang, and T. B. Hashimoto, "Alpaca: A strong, replicable instruction-following model," Stanford CRFM Blog, 2023.

[3] M. Conover, M. Hayes, A. Mathur, J. Xie, J. Wan, S. Shah, A. Ghodsi, P. Wendell, M. Zaharia, and R. Xin, "Free dolly: Introducing the world's first truly open instruction-tuned LLM," https://www.databricks.com/blog/2023/04/12/dolly-first-open-commercially-viable-instruction-tuned-llm, 2023.